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Python Control Statements
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** Control statements are used to control the flow of execution in a Python program.

** Normally, Python executes statements from top to bottom.

** But in real-time applications, we need to execute code based on conditions or repeat some code multiple times.
```

Ex:

```
if(uname == "admin" and pwd == "admin123"){
    // logic to display dashboard page
}else{
    //logic to display error msg
}
```

Why Do We Need Control Statements?

```
** In real-time applications, we need decision-making and repetition.
```

Real-Time Examples

- 1) If user logged in, show dashboard page
- 2) If student marks are greater than 35, show pass
- 3) If account balance is sufficient, allow withdrawal / transfer
- 4) Print numbers from 1 to 100
- 5) Send certificates to all students
- 6) Display all products in cart
- 7) Search a student from a list
- 8) Repeat login until correct password is entered

```
** Without control statements, programs cannot make decisions or repeat tasks.
```

```
*** Python control statements are mainly divided into three types:
```

1. Conditional Statements
2. Looping Statements
3. Jumping Statements (loop control statements)

```
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Conditional Statements
```

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=> Conditional statements are used to execute the logic based on condition

=> In conditional statements the logic will execute only once.

if statement

** The if statement is used to execute a block of code only when the condition is true.

syntax :

```
if condition:
    statement
```

Example

```
age = 25
```

```
if age>=18:
    print("Eligible to vote")
    print("plz participate in voting")
```

Indentation in Python

** Python uses indentation to define blocks of code.

** Indentation means spaces at the beginning of a line.

wrong syntax

```
age = 20
```

```
if age >= 18:
    print("Eligible to vote")
```

if-else Statement

syntax :

```
if condition:
    statement1
else:
    statement2
```

Example

```
age = 12
```

```
if age>=18:
    print("Eligible to vote")
    print("plz participate in voting")

else:
    print("Not eligible to vote")
```

if-elif-else statements

**** When we want to check multiple conditions like below then we need to use if-elif-else concept.**

```
marks >= 90 (A Grade)
```

```
marks >= 75 (B grade)
```

```
marks >= 60 (C grade)
```

```
marks >= 35 (Pass)
```

```
else ==> Fail
```

Syntax :

```
if condition1:
    statement1
elif condition2:
    statement2
elif condition3:
    statement3
else:
    default_statement
```

Example

```
marks = int(input("Enter marks: "))
```

```
if marks >= 90:
    print("A Grade")
elif marks >= 75:
    print("B Grade")
elif marks >= 65:
    print("C Grade")
elif marks >= 35:
    print("Just Pass")
else:
    print("Failed")
```

Nested If stmt

**** Writing one if statement inside another if statement is called as nested if.**

Example-1 ::

```
if username == "admin" and pwd == "admin123":
```

```
    if role == "student":
        print("student dashboard")
```

```
    elif role == "trainer":
        print("trainer dashboard")
```

```
    elif role == "admin":
        print("admin dashboard")
```

```
else:
    print("Invalid Credentials")
```

Example-2 ::

```
balance = 5000
withdraw_amount = int(input("Enter amount to withdraw: "))

if balance > 0 :
    if withdraw_amount <= balance:
        balance = balance - withdraw_amount
        print("Withdraw successful")
        print("Remaining balance: ", balance)
    else:
        print("Insufficient balance")
else:
    print("Funds not available")

##### match case #####

print("1. Add Student")
print("2. View Student")
print("3. Update Student")
print("4. Delete Student")

choice = int(input("Enter your choice: "))

match choice:
    case 1:
        print("Student add operation selected")
    case 2:
        print("Student View Operation selected")
    case 3:
        print("Student Update Operation selected")
    case 4:
        print("Student Delete Operation selected")
    case _:
        print("Invalid Choice")
```